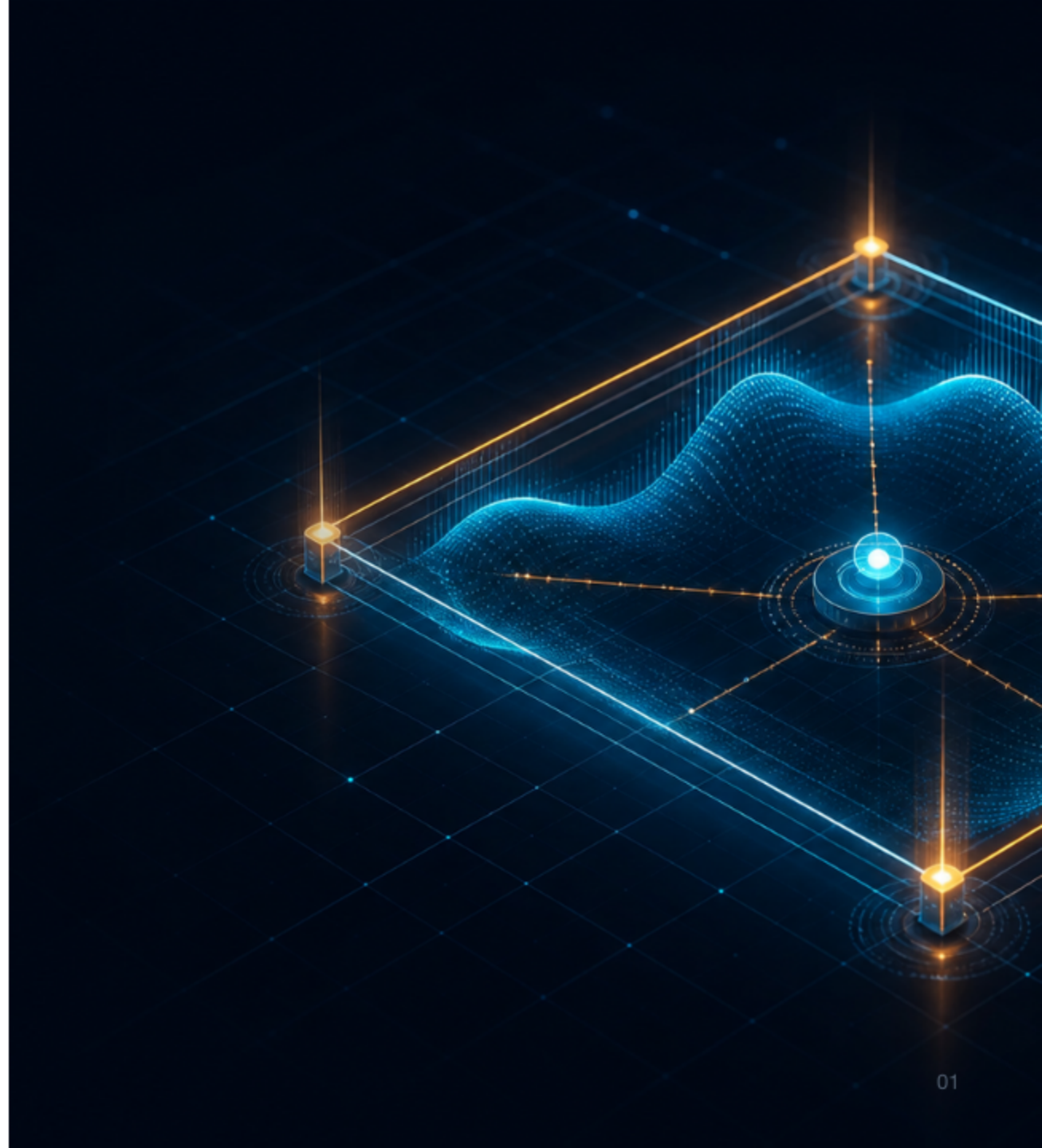


ThetaTrap

A bounded AI agent for paper options trading

Qwen makes a real tool decision through Alpaca's official MCP server. Deterministic policy controls every number and risk limit.

RYO KITANO / PAPER TRADING / BASIC INDICATIVE DATA



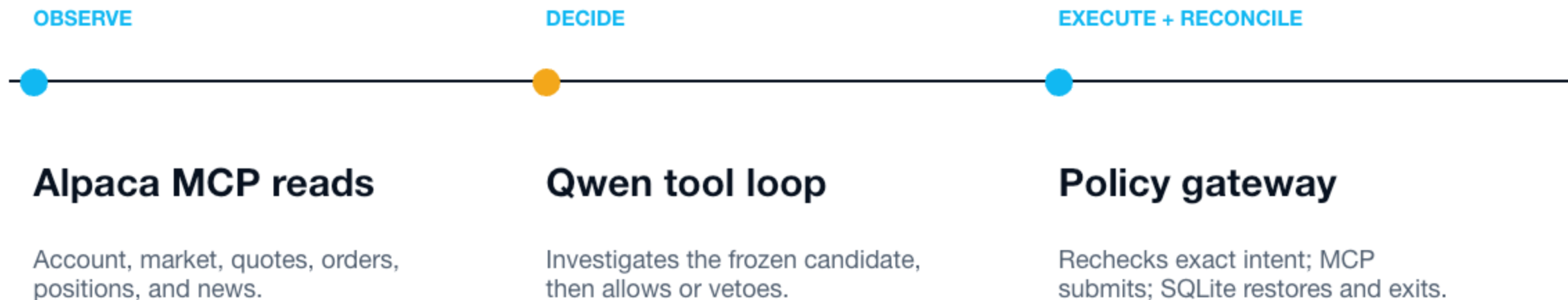
One simple idea. One difficult execution problem.

**This was my first options project.
A YouTube lesson made me ask:
can AI trade expensive earnings
premium with defined risk?**

An iron condor caps loss, but it still needs four usable quotes, enough liquidity, and reliable multi-leg execution. Missing evidence must mean no trade.

- | | | |
|----|------------------|--|
| 01 | Idea | Earnings can make option premium expensive. |
| 02 | Structure | Use one contract, four legs, and a known maximum loss. |
| 03 | Reality | Basic indicative data often lacked complete four-leg evidence. |
| 04 | Adapt | Keep the boundaries while testing one date-scoped ETF canary. |

AI judgment inside deterministic risk.



Hard boundary: Qwen cannot change the symbol, strikes, quantity, price, loss cap, account, or authorization.

The final canary was small, bounded, and one-shot.

After 314 rejected earnings evaluations, the Sep 3 adaptation tested the same Qwen, MCP, policy, and recovery workflow on QQQ and SPY.

\$80

Loss cap

The selected QQQ structure calculated \$18 maximum loss.

1

Contract

One four-leg position; no naked options.

1

Broker entry

SPY was blocked before mutation; QQQ used the only order.

One complete round trip. One honest result.

LIVE COMPETITION AUDIT - BUILD 9CDAB8519961

ThetaTrap

A bounded MCP-native options agent. The original earnings strategy stayed fail-closed when Alpaca Basic data could not support a valid four-leg trade; the date-scoped Sep 3 Intraday Theta Canary is the transparent, testable response.

PAPER TRADINGALPACA OFFICIAL MCPPUBLIC READ-ONLY VIEWBASIC INDICATIVE DATA

00 - RESULT SNAPSHOT

The outcome, before the implementation details

Current broker and agent evidence from the competition account. Values refresh every 30 seconds.

PAPER - TRADING DISARMED

Final run state
FLAT

Position
FLAT

Starting equity
\$100,000.00

Latest equity
\$99,986.80

Observed P&L
-\$13.20

Qwen result
ALLOW

Worker HEALTHY - MCP READY - market CLOSED - heartbeat 2026-09-03T22:06:40.932601+00:00 (30s old) - last successful MCP call 2026-09-03T22:06:40.857135+00:00

01 - PROBLEM & STRATEGY

Sell bounded premium only when the evidence is usable

ThetaTrap started with earnings volatility, then introduced one transparent Sep 3 canary after the Basic feed repeatedly failed complete four-leg gates.

QQQ

PAPER TRADE

\$0.82 credit in; \$0.95 debit out.

-\$13.20

ACCOUNT P&L

Final equity: \$99,986.80 from \$100,000.00.

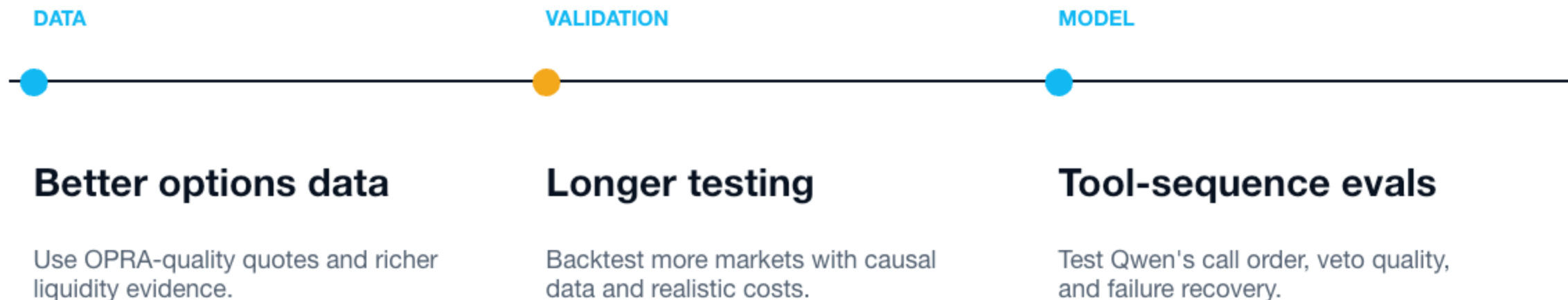
FLAT

END STATE

Confirmed by 3:17 PM ET; zero open orders.

10 tool calls · 7 via Alpaca MCP · Qwen: ALLOW

The workflow worked. Profitability is still unproven.



My first options project became a complete agent. Next: make the evidence deeper and the trading system more robust.